

NE504 - Nuclear fuel cycle analysis

Xenon neutron poison production

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1 Mathematical models

1.1 Iodine

Rate of change of iodine -

$$\begin{aligned}\frac{dI}{dt} &= \gamma_I \Sigma_F \phi_T - \lambda_I I \\ I(0) &= 0\end{aligned}\tag{1}$$

1.2 Xenon

Rate of change of xenon -

$$\begin{aligned}\frac{dX}{dt} &= \lambda_I I + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \\ X(0) &= 0\end{aligned}\tag{2}$$

2 Solutions

2.1 Iodine

Apply laplace transform to Eq. (51) -

$$\begin{aligned}\frac{dI}{dt} &= \gamma_I \Sigma_F \phi_T - \lambda_I I \\ s\tilde{I} &= \frac{1}{s} \gamma_I \Sigma_F \phi_T - \lambda_I \tilde{I}\end{aligned}\tag{3}$$

Rearrange terms to obtain laplace solution -

$$\tilde{I} = \frac{1}{s(s + \lambda_I)} \gamma_I \Sigma_F \phi_T\tag{4}$$

Invert Eq. (4) to obtain real time domain solution -

$$I(t) = \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} (1 - e^{-\lambda_I t})\tag{5}$$

2.2 Xenon

Substitute the iodine solution in Eq. (14) into Eq. (53) -

$$\begin{aligned}\frac{dX}{dt} &= \lambda_I I + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \\ \frac{dX}{dt} &= \lambda_I \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} (1 - e^{-\lambda_I t}) + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X\end{aligned}\tag{6}$$

Rearrange the terms and simplify -

$$\begin{aligned}\frac{dX}{dt} &= \gamma_I \Sigma_F \phi_T (1 - e^{-\lambda_I t}) + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \\ \frac{dX}{dt} &= \gamma_I \Sigma_F \phi_T - \gamma_I \Sigma_F \phi_T e^{-\lambda_I t} + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X\end{aligned}\tag{7}$$

$$\frac{dX}{dt} = \gamma_I \Sigma_F \phi_T - \gamma_I \Sigma_F \phi_T e^{-\lambda_I t} + \gamma_X \Sigma_F \phi_T - (\lambda_X + \sigma_A \phi_T) X \quad (8)$$

Apply laplace transform to Eq. (8) -

$$s\tilde{X} = \frac{1}{s} \gamma_I \Sigma_F \phi_T - \frac{1}{s + \lambda_I} \gamma_I \Sigma_F \phi_T + \frac{1}{s} \gamma_X \Sigma_F \phi_T - (\lambda_X + \sigma_A \phi_T) \tilde{X} \quad (9)$$

Rearrange the terms in Eq. (9) to obtain the laplace solution -

$$(s + [\lambda_X + \sigma_A \phi_T]) \tilde{X} = \frac{1}{s} \gamma_I \Sigma_F \phi_T - \frac{1}{s + \lambda_I} \gamma_I \Sigma_F \phi_T + \frac{1}{s} \gamma_X \Sigma_F \phi_T \quad (10)$$

Then -

$$\tilde{X} = + \frac{1}{s(s + [\lambda_X + \sigma_A \phi_T])} \gamma_I \Sigma_F \phi_T - \frac{1}{(s + \lambda_I)(s + [\lambda_X + \sigma_A \phi_T])} \gamma_I \Sigma_F \phi_T + \frac{1}{s(s + [\lambda_X + \sigma_A \phi_T])} \gamma_X \Sigma_F \phi_T \quad (11)$$

Invert Eq. (11) to obtain real time domain solution -

$$X(t) = \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} (1 - e^{-(\lambda_X + \sigma_A \phi_T)t}) - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} (e^{-\lambda_I t} - e^{-(\lambda_X + \sigma_A \phi_T)t}) + \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} (1 - e^{-(\lambda_X + \sigma_A \phi_T)t}) \quad (12)$$

3 Derivatives

3.1 Iodine

$$I(t) = \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} (1 - e^{-\lambda_I t}) \quad (13)$$

Expand -

$$I(t) = \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} e^{-\lambda_I t} \quad (14)$$

Compute derivative -

$$\begin{aligned} \frac{dI}{dt} &= \frac{d}{dt} \left[\frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} \right] - \frac{d}{dt} \left[\frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} e^{-\lambda_I t} \right] \\ \frac{dI}{dt} &= 0 - (-\lambda_I) \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} e^{-\lambda_I t} \right) \end{aligned} \quad (15)$$

Therefore -

$$\frac{dI}{dt} = \gamma_I \Sigma_F \phi_T e^{-\lambda_I t} \quad (16)$$

3.2 Xenon

$$X(t) = \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} (1 - e^{-(\lambda_X + \sigma_A \phi_T)t}) - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} (e^{-\lambda_I t} - e^{-(\lambda_X + \sigma_A \phi_T)t}) + \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} (1 - e^{-(\lambda_X + \sigma_A \phi_T)t}) \quad (17)$$

Rearrange to make computing the derivative easier -

$$\begin{aligned} X(t) &= \\ &\left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} \right) \\ &- \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-(\lambda_X + \sigma_A \phi_T)t} \right) \\ &+ \left(\frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} \right) \end{aligned} \quad (18)$$

$$\begin{aligned}
X(t) = & \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} \right) \\
& - \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-(\lambda_X + \sigma_A \phi_T)t} \right) \\
& + \left(\frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} \right)
\end{aligned} \tag{19}$$

$$\begin{aligned}
X(t) = & \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} \\
& - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-(\lambda_X + \sigma_A \phi_T)t} \\
& + \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t}
\end{aligned} \tag{20}$$

Group the terms by exponent -

$$\begin{aligned}
X(t) = & \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} + \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \\
& - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} \\
& + \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-(\lambda_X + \sigma_A \phi_T)t} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} - \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t}
\end{aligned} \tag{21}$$

$$X(t) = \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-(\lambda_X + \sigma_A \phi_T)t} - \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} e^{-(\lambda_X + \sigma_A \phi_T)t} \tag{22}$$

$$X(t) = \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right) e^{-(\lambda_X + \sigma_A \phi_T)t} \tag{23}$$

Now compute the derivative -

$$\frac{dX}{dt} = \frac{d}{dt} \left[\frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right] - \frac{d}{dt} \left[\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} \right] + \frac{d}{dt} \left[\left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \right] \quad (24)$$

$$\frac{dX}{dt} = 0 + (\lambda_I) \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - (\lambda_X + \sigma_A \phi_T) \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (25)$$

$$\frac{dX}{dt} = +(\lambda_I) \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - (\lambda_X + \sigma_A \phi_T) \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (26)$$

$$\frac{dX}{dt} = \frac{\gamma_I \lambda_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - (\lambda_X + \sigma_A \phi_T) \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (27)$$

$$\frac{dX}{dt} = \frac{\gamma_I \lambda_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{(\lambda_X + \sigma_A \phi_T) (\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (28)$$

$$\frac{dX}{dt} = \frac{\gamma_I \lambda_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \Sigma_F \phi_T \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (29)$$

$$\frac{dX}{dt} = \frac{\gamma_I \lambda_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \Sigma_F \phi_T \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (30)$$

$$\frac{dX}{dt} = \Sigma_F \phi_T \frac{\gamma_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \Sigma_F \phi_T \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (31)$$

4 Equilibrium time

The equilibrium time for xenon should be when the concentration achieves a steady state; i.e., the rate of change in the concentration is zero.

$$\frac{dI}{dt} = \gamma_I \Sigma_F \phi_T e^{-\lambda_I t} \quad (32)$$

$$\frac{dX}{dt} = \Sigma_F \phi_T \frac{\gamma_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \Sigma_F \phi_T \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (33)$$

4.1 Iodine

$$\gamma_I \Sigma_F \phi_T e^{-\lambda_I t} = 0 \quad (34)$$

Here, the constants drop out -

$$e^{-\lambda_I t} = 0 \quad (35)$$

4.2 Xenon

$$\Sigma_F \phi_T \frac{\gamma_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \Sigma_F \phi_T \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} = 0 \quad (36)$$

With this equation, $\Sigma_F \phi_T$ drops out -

$$\frac{\gamma_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} = 0 \quad (37)$$

4.3 Solution

However, neither equation for iodine or xenon cannot be solved explicitly for t because $\ln[0]$ is not defined. There are a variety of techniques that can be applied to obtain the equilibrium time.

5 Approximation methods

5.1 Newton's method

For $f(t) = 0$ -

$$t_{n+1} = t_n - \frac{f(t_n)}{f'(t_n)}$$

5.1.1 Iodine

$$e^{-\lambda_I t} = 0 \quad (38)$$

$$f_I(t) = e^{-\lambda_I t} \quad (39)$$

$$f'_I(t) = -\lambda_I e^{-\lambda_I t} \quad (40)$$

5.1.2 Xenon

$$\frac{\gamma_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} = 0 \quad (41)$$

$$f_X(t) = \frac{\gamma_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} - \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (42)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + (\lambda_X + \sigma_A \phi_T) \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - (\gamma_I + \gamma_X) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (43)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + (\lambda_X + \sigma_A \phi_T) \left(\frac{(\lambda_X + \sigma_A \phi_T) \gamma_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \gamma_I - \gamma_X \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (44)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + (\lambda_X + \sigma_A \phi_T) \left(\frac{\gamma_I (\lambda_X + \sigma_A \phi_T)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{\gamma_I (\lambda_X + \sigma_A \phi_T - \lambda_I)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \gamma_X \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (45)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \left(\frac{\gamma_I (\lambda_X + \sigma_A \phi_T) (\lambda_X + \sigma_A \phi_T)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \frac{\gamma_I (\lambda_X + \sigma_A \phi_T) (\lambda_X + \sigma_A \phi_T - \lambda_I)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \gamma_X (\lambda_X + \sigma_A \phi_T) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (46)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \left(\frac{\gamma_I (\lambda_X + \sigma_A \phi_T) (\lambda_X + \sigma_A \phi_T) - \gamma_I (\lambda_X + \sigma_A \phi_T) (\lambda_X + \sigma_A \phi_T - \lambda_I)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \gamma_X (\lambda_X + \sigma_A \phi_T) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (47)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \left(\frac{\gamma_I (\lambda_X + \sigma_A \phi_T) (\lambda_X + \sigma_A \phi_T - \lambda_X - \sigma_A \phi_T + \lambda_I)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \gamma_X (\lambda_X + \sigma_A \phi_T) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (48)$$

$$f'_X(t) = -\frac{\gamma_I \lambda_I \lambda_I}{\lambda_X + \sigma_A \phi_T - \lambda_I} e^{-\lambda_I t} + \left(\frac{\gamma_I \lambda_I (\lambda_X + \sigma_A \phi_T)}{\lambda_X + \sigma_A \phi_T - \lambda_I} - \gamma_X (\lambda_X + \sigma_A \phi_T) \right) e^{-(\lambda_X + \sigma_A \phi_T) t} \quad (49)$$

6 Steady state concentrations

The concentration of I and Xe at steady state can be obtained by placing the derivatives equal to zero; $\frac{dI}{dt} = 0$ and $\frac{dX}{dt} = 0$.

Iodine -

$$\frac{dI}{dt} = \gamma_I \Sigma_F \phi_T - \lambda_I I \quad (50)$$

$$\begin{aligned} \gamma_I \Sigma_F \phi_T - \lambda_I I &= 0 \\ \lambda_I I &= \gamma_I \Sigma_F \phi_T \\ I_\infty &= \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} \end{aligned} \quad (51)$$

$$\frac{dX}{dt} = \lambda_I I + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \quad (52)$$

$$\begin{aligned} \lambda_I I + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X &= 0 \\ \lambda_I \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} \right) + \gamma_X \Sigma_F \phi_T - (\lambda_X + \sigma_A \phi_T) X &= 0 \\ (\lambda_X + \sigma_A \phi_T) X &= \lambda_I \left(\frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} \right) + \gamma_X \Sigma_F \phi_T \\ (\lambda_X + \sigma_A \phi_T) X &= \gamma_I \Sigma_F \phi_T + \gamma_X \Sigma_F \phi_T \\ X_\infty &= \frac{(\gamma_I + \gamma_X) \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \end{aligned} \quad (53)$$